Applied Cryptography

Mandatory assignment two: Chat application v2

Table of Contents

Introduction 2
Requirements: 2
Technologies suggestion 2
Rules and Deadline 3
What you're asked to do 3

Introduction

Development of Python Chat application with the use of Implementing a client-server application using **AES-GCM (Advanced Encryption Standard in Galois/Counter Mode)** that provides both confidentiality (encryption) and integrity (authentication) strategy based on TCP/IP client/server design pattern.

There are two types of architectures for solving the assignment: See figure 1 and 2

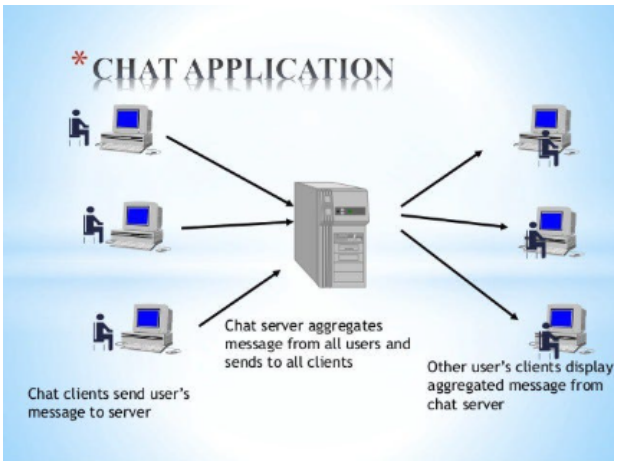

Figure 1, Chat application with central server

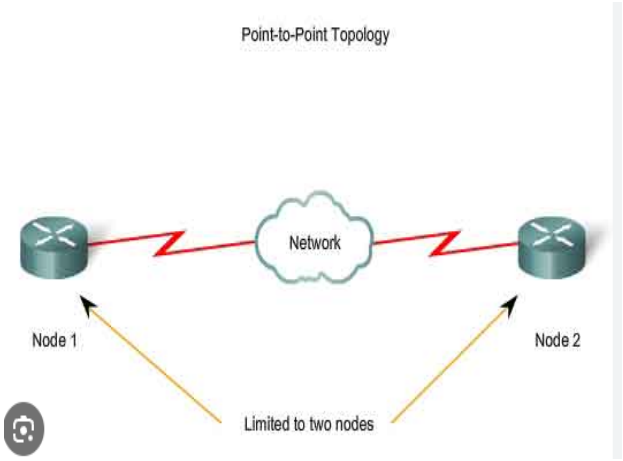

Figure 2, Char application with point-to-point communication

Requirements:

Development of solution one demands two files -> server.py and client.py

Development of solution two demands two files -> client_1.py and client_2.py

Key Exchange between clients for AES-GCM: Use one-time key exchange over UDP/IP according to figure 3

Use input() function for chat communication (commando prompt)

Technologies suggestion:

```
from cryptography.hazmat.primitives.ciphers.aead import AESGCM
```

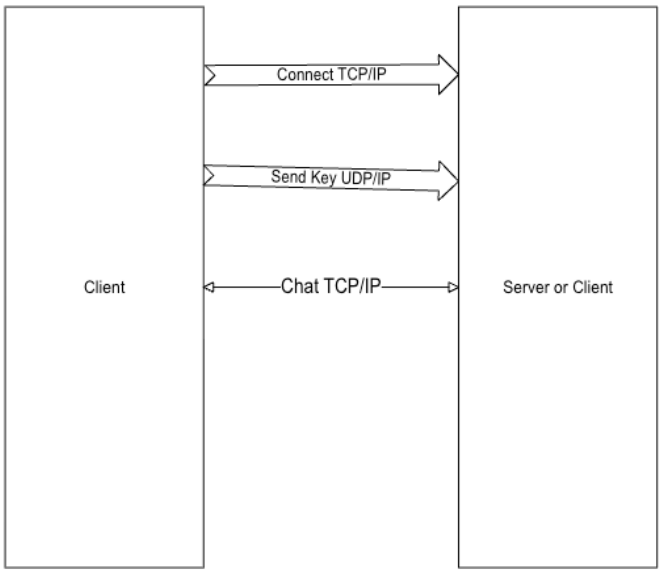

Figure 3, Application communication TCP/IP with AES-GCM and key exchange with UDP/IP

Rules and Deadline

This assignment is expected to be made in **groups of 1-2 students**.

Hand in of mandatory: 29/04-2026 before 23:59

What you're asked to do

1. Make the application in python
2. Test it with Wireshark to see the ciphertext
3. Make documentation with focus on the cryptography implementation and the test with Wireshark.
    a. 1-3 pages max in pdf format. Normal number of characters per page.
    b. Title and name(s)
    c. Cryptography implementation
    d. There must be a wireshark screendump of the ciphertext data in a TCP data field
    e. There must be a wireshark screendump of the ciphertext/cleartext data in a UDP data field
    f. How to use it
4. Summit a **zip** file without password as xxxxx.zip or xxxxx_xxxxx.zip where xxxxx is the EK username. The zip file must have two py files and one pdf file.